SLEEPNG STAGES DETECTION USING EOG CHANNELS

SP
Signal Processing

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Signal Processing Course
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1st Abstract:

In this report, we will explain what we did in the Sleep Stages Classification project for the Signal Processing Course.

The aim of this project was to determine whether we could classify sleeping stages **(Wake, N1(Light Sleep), N2(Intermediate Sleep), N3(Deep Sleep), REM**) with a single channel only (**except the EEG**) and the answer is **YES using EOG**, we can!

We used the MESA (Multi-Ethnic Study of Atherosclerosis) Sleep Dataset which is used to study sleep patterns and sleep disorders.

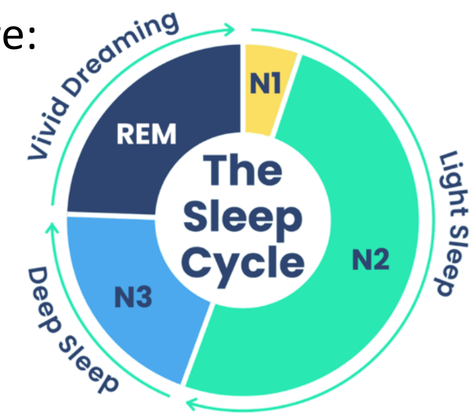
Number of participants In the MESA Sleep Dataset is: 2237, we chose only 10 participants for our project


```
Loaded: mesa-sleep-0001.edf — 1439 epochs → 1429 sequences
Loaded: mesa-sleep-0002.edf — 1319 epochs → 1309 sequences
Loaded: mesa-sleep-0006.edf — 1079 epochs → 1069 sequences
Loaded: mesa-sleep-0010.edf — 1199 epochs → 1189 sequences
Loaded: mesa-sleep-0012.edf — 1427 epochs → 1417 sequences
Loaded: mesa-sleep-0014.edf — 1679 epochs → 1669 sequences
Loaded: mesa-sleep-0016.edf — 1199 epochs → 1189 sequences
Loaded: mesa-sleep-0021.edf — 1079 epochs → 1069 sequences
Loaded: mesa-sleep-0027.edf — 1199 epochs → 1189 sequences
Loaded: mesa-sleep-0028.edf — 1139 epochs → 1129 sequences

Total subjects loaded: 10
```

2nd Introduction:

The human health and well-being require sufficient sleep. Sleep disorders (sleep apnea, insomnia and quality of sleep) are correlated with cardiovascular disease, diabetes, and cognitive impairment. The sleep staging analysis has limitations as the manual scoring by sleep specialist requires time consuming and expensive. However, classification of sleep stages can be automatically performed by using deep learning and the analysis becomes time and cost efficient.

Sleep staging is the classification of sleep into various stages using physiological data, The main sleep stages are:



- Wake (W)
- N1 (Light sleep)
- N2 (Intermediate sleep)
- N3 (Deep sleep)
- REM (Rapid Eye Movement sleep)

Sleep stages are typically scored into 30 second epochs using American Academy of Sleep Medicine (AASM) scoring guidelines.

Sleep stages can be used for:

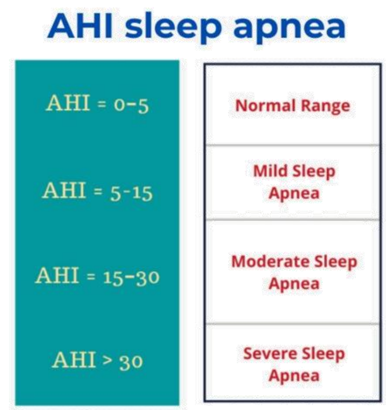
- Diagnosing sleep disorders
- assessing the quality of sleep
- monitoring patient well-being
- learning about cardiovascular and neurological disorders

Apnea-Hypopnea Index (AHI):

From The Vivos Institute & Cleveland Clinic:

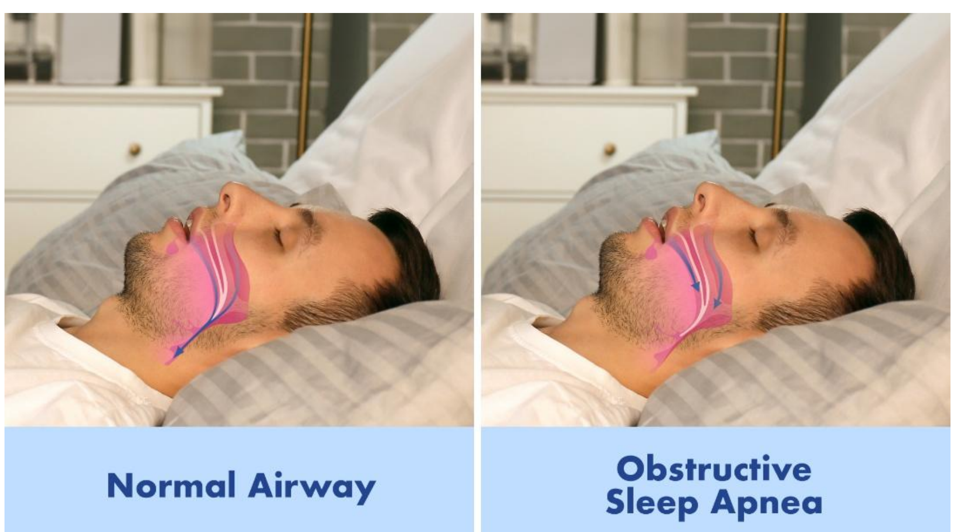
The Apnea-Hypopnea Index (AHI) is considered to be the single most important indicator of sleep quality. The number of times, on average, that each person stops breathing (apnea) or dramatically reduces breathing (hypopnea) at least 10 times per hour of sleep. AHI is the diagnostic test for doctors diagnosing Obstructive Sleep Apnea (OSA).

The AHI can be interpreted by the adult as the following sleep disturbances:



- Normal: <5 events/ hr.
- Mild SA: 5-14 events/hr.
- Moderate SA: 15-29 events/hr.
- Severe SA: 30+ events/hr.

For a child the AHI should be < 1 otherwise it can be considered abnormal. An AHI > 5 is often considered severe in children due to their less compliant airways.



AHI effect on sleep quality:

- **Reduced quality of sleep:** High AHI indicates frequent mini-arousals, the brain forcing the person to awake just long enough to resume breathing, this thus interrupts deeper (and vital for health) sleep, and REM sleep.
- **Oxygen Desaturation**: The recurring events each bring about an increase in blood oxygen level drop (oxygen desaturation), a metabolic stressor.
- **Daytime problems**: High AHI indicates increase in daytime sleepiness, headaches upon waking in the morning, and poorer mental functioning.

Factors affecting AHI:

- **Sleep Position**: AHI is typically increased in the back sleeping position (supine) secondary to the effect of gravity on the airway, while in side sleeping position it may be decreased.
- **Weight**: As BMI (Body Mass Index) is increased the airway is subjected to greater pressure and the AHI is increased.
- **Sex**: The average AHI is higher in the male
- **Method of Measurement**: In the event of using a HST (Home Sleep Tests) a reduced AHI may be observed compared to a PSG in an inpatient facility, due to calculating the events as per the total recording time as opposed to total sleeping time.

3rd Dataset:

MESA Sleep Dataset is a huge biomedical dataset that we are going to analyze to understand sleep patterns, sleep disorders, etc., MESA stands for: Multi-Ethnic Study of Atherosclerosis

This was part of a study for cardiovascular disease research.

Some features of the MESA dataset are as follows:

- Number of participants: 2237 (We are going to work on 10 Individuals only)
- Range of participants ages: 45 to 84 years
- Multi-ethnic people in the study: Black - White - Hispanic - Chinese
- Overnight recordings during polysomnography.

There are various kinds of data available in this dataset:

- Polysomnography (PSG) signals
- Sleep stage annotations
- Actigraphy recordings
- Clinical & demographic data
- Sleep questionnaires

For this project, we will use the PSG signals and the sleep stage annotations.

Dataset Link

PSG records multiple physiological signals including:

EEG (Electroencephalogram) → brain activity
ECG (Electrocardiogram) → heart activity
EMG (Electromyography) → muscle activity
EOG (Electrooculogram) → eye movement
SpO_2 → oxygen saturation
Respiratory signals

4th Classes:

They are the class names (classifications) indicating the stage of the recorded sleep signal in a sleep study.

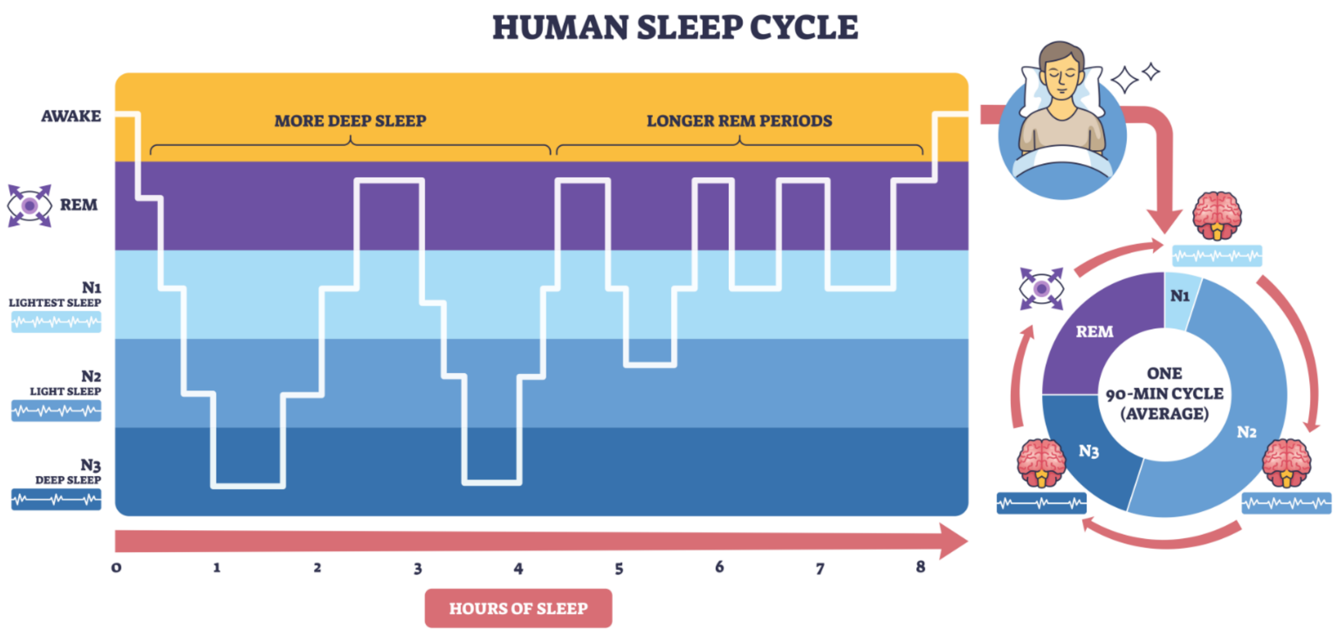
A Polysomnography (PSG) study involves recording the physiological signals such as the EEG, EOG and EMG overnight. The signals are then examined by a sleep expert who classifies every epoch (a short time period of sleep) into a specific sleep stage.

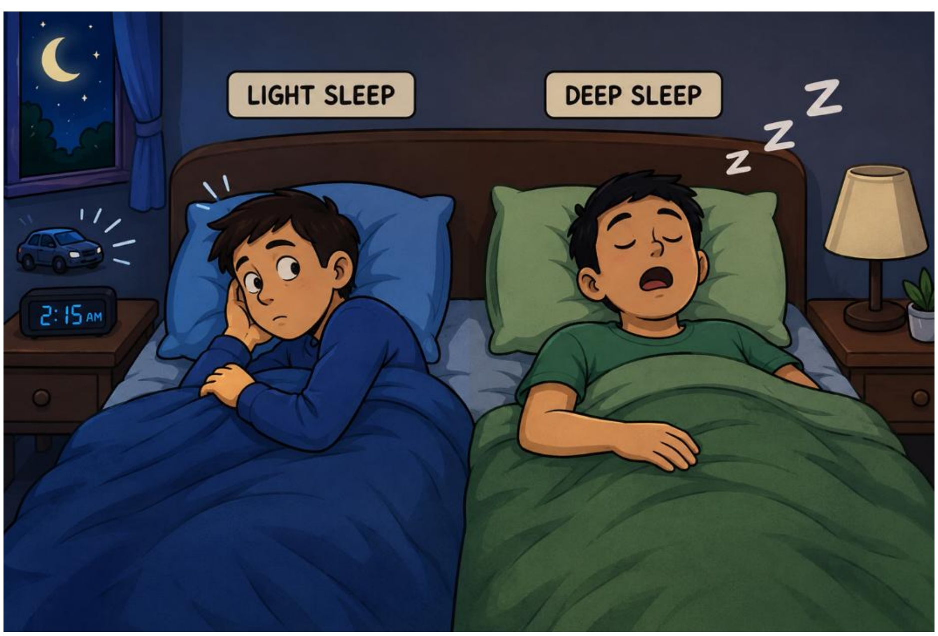
Typically, each epoch lasts for 30 seconds worth of sleep.

Under the AASM (American Academy of Sleep Medicine) scoring system, sleep is categorized as the following stages:

- Wake (W): the patient is awake.
- N1: the patient is sleeping lightly and is undergoing a period where the patient is in between awake and asleep.
- N2: a stage of sleep where the patient has moved into a deeper sleep where the heart and body temperature decreases.

- N3: this is deep sleep and slow-wave sleep which is essential for bodily recovery.
- REM: Rapid eye movement sleep is the stage of sleep where the patient dreams and when the electrical activity in the brain resembles a sleeping brain.

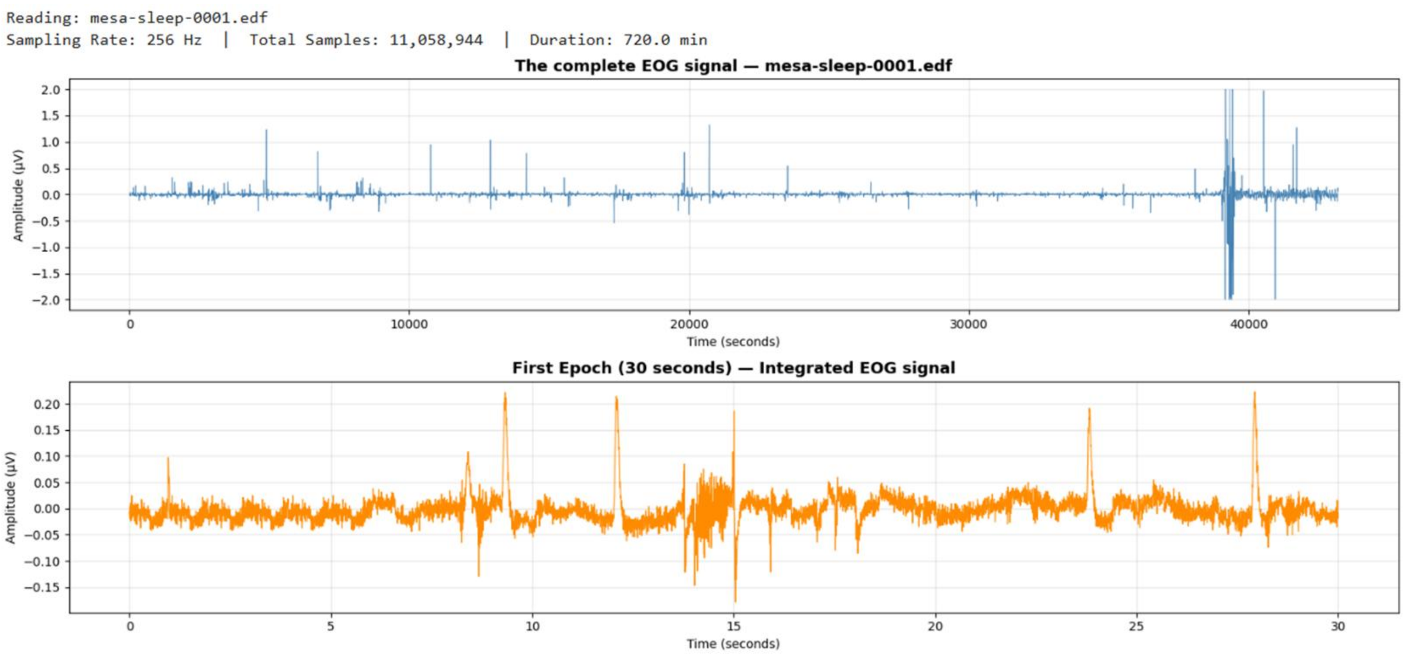


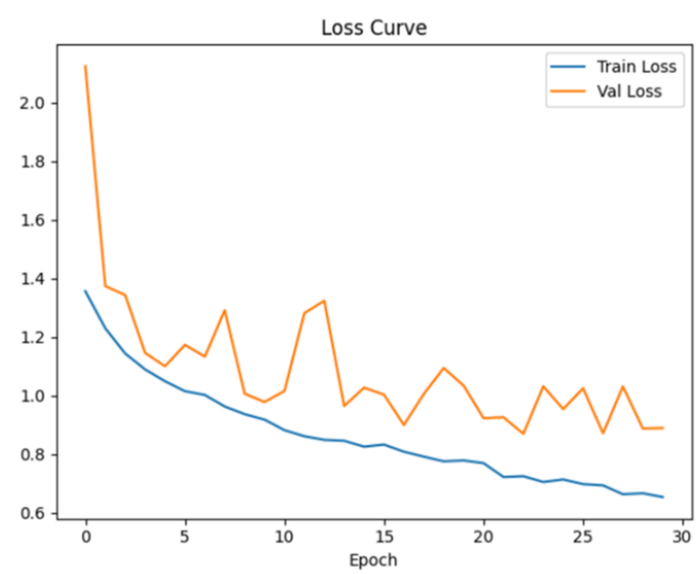


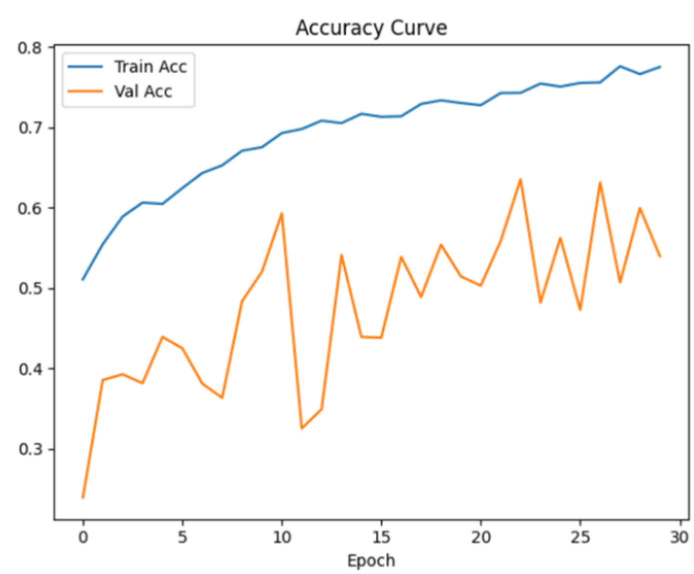
5th Our Work:

We chose the EOG Signal for our project to classify sleeping stages.

Firstly, we worked on the average of both left and right EOG signals to prevent having NULL values at some instants and these are the results we got:

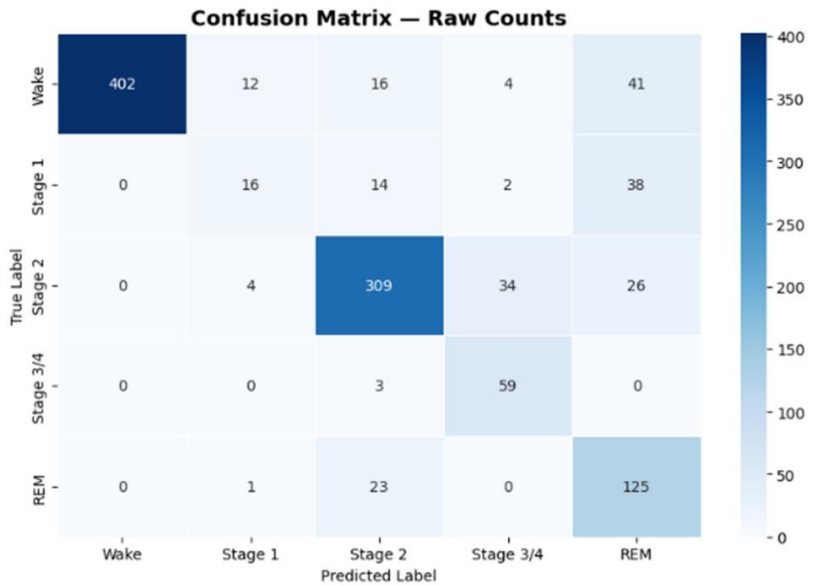






	precision	recall	f1-score	support
Wake	1.00	0.85	0.92	475
Stage 1	0.48	0.23	0.31	70
Stage 2	0.85	0.83	0.84	373
Stage 3/4	0.60	0.95	0.73	62
REM	0.54	0.84	0.66	149
accuracy			0.81	1129
macro avg	0.69	0.74	0.69	1129
weighted avg	0.83	0.81	0.81	1129

Macro F1 : 0.6915
Cohen Kappa: 0.728



Then we worked only on a single channel as requested, we worked one time on left eye and the other on right eye , our main project and the best results we got was on the right EOG but here we will mention each and every step we took to get the best possible outcome , we worked on multiple notebooks and got different results by making some changes every time which we will mention moving forward in this report.

General steps for both left & right:

1. We loaded the EDF files.
2. Reading the EOG Signal
3. Bandpass Filtering (0.3-35 Hz) using Butterworth filter to remove noise artifacts and get desired frequencies
4. Creating a 30 Seconds Epoch
5. Load XML Labels
6. Normalization (Standard Deviation =1 & Mean= 0)
7. Creating a sequence of 10 epochs as the sleeping stages are corelated

8. Subject Wise Split to prevent data leakage
9. Creating a CNN+LSTM model
10. Classification of classes (making a better format so it can be used by the model properly)
11. Training the model with early stopping to save memory
12. Evaluation & Plotting

Firstly, we tried on both left and right with same model and splitting **(70-20-10)** using subject split, the right eye model gave us better results (accuracy 71%) while left was (accuracy 69%) so we directed ourself to work on the **EOG-R**

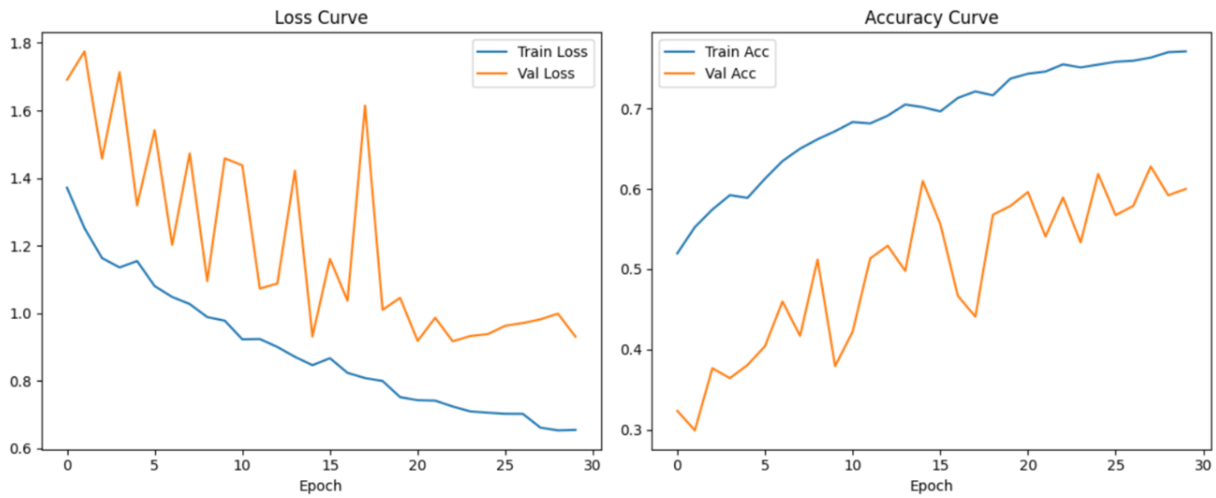
	precision	recall	f1-score	support
Wake	1.00	0.71	0.83	475
Stage 1	0.20	0.41	0.27	70
Stage 2	0.88	0.70	0.78	373
Stage 3/4	0.39	1.00	0.56	62
REM	0.62	0.79	0.69	149
accuracy			0.71	1129
macro avg	0.62	0.72	0.63	1129
weighted avg	0.82	0.71	0.75	1129

Macro F1 : 0.6262
Cohen Kappa: 0.619

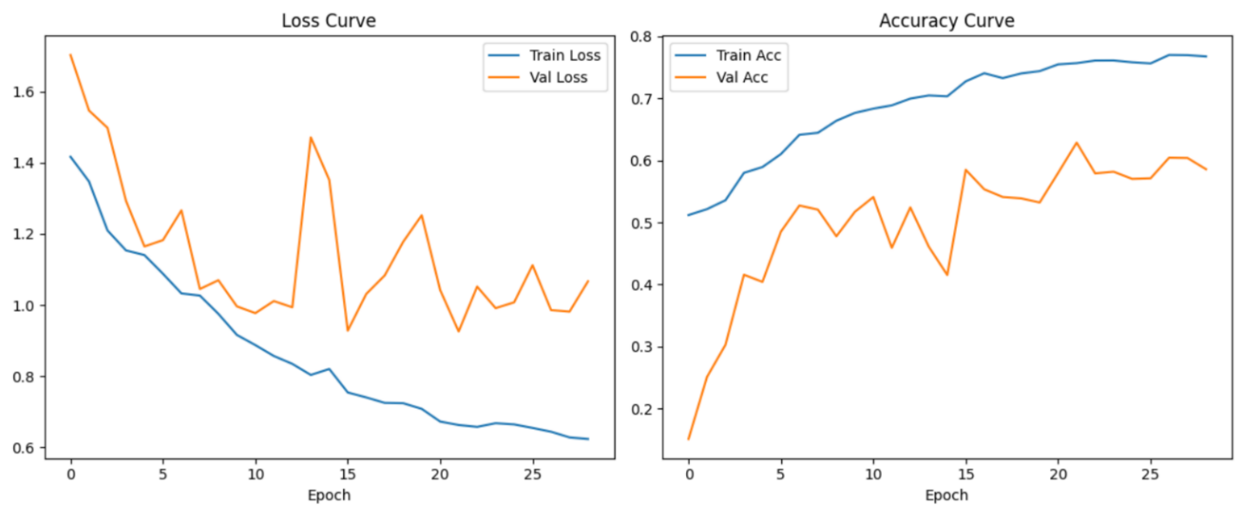
EOG-L	precision	recall	f1-score	support
Wake	1.00	0.84	0.91	475
Stage 1	0.42	0.24	0.31	70
Stage 2	0.79	0.50	0.61	373
Stage 3/4	0.83	0.48	0.61	62
REM	0.35	0.98	0.52	149
accuracy			0.69	1129
macro avg	0.68	0.61	0.59	1129
weighted avg	0.80	0.69	0.71	1129

Macro F1 : 0.5937
Cohen Kappa: 0.5768

EOG-R



EOG-L



Here we worked on several notebooks:

1. P_EOG-R (1) (It was mentioned above)

Main Characteristics

It seems that it is an earlier stable release.

Key Features

- EOG Right channel was utilized only.

- Deep learning architecture consists of:
  - CNN Encoder
  - LSTM
- StandardScaler normalization
- CrossEntropyLoss
- ReduceLROnPlateau scheduler
- Metrics utilized:
  - Accuracy
  - F1-score
  - Cohen Kappa

Comments

This notebook served as the base for the pipeline:

- Preprocessing
- Sequence creation
- Model training
- Cvaluation

Possible shortcomings

- The subjects split could not be improved yet.
- The class imbalance could be improved.
- Generalization across subjects was not perfect yet.

2. P_EOG-L (It was mentioned above)

Main Difference

Here is the change from:

- Right EOG channel

to:

- Left EOG channel

Why this experiment was likely done:

This experiment could have been done in order to:

- Compare EOG-L versus EOG-R performance
- Test the difference in quality between signals from different leads
- See if the channel selection matters, i.e. How channel robustness compares

Scientific Significance:

This type of comparison would aid in:

- Identifying the EOG lead which carries the most discriminative information
- Concluding whether performance at sleep stage classification differs when using the Left EOG lead compared to the Right EOG lead

Later we decided to do the **EOG-R**, so we did multiple modifications on multiple notebooks:

1.P_EOG-R-subject-split (2)

Among the Most Critical Enhancements

This notebook was notable for adding a superior splitting method on a subject level.

Significant Enhancement

Subject-Based Splitting

This notebook focuses on:

- aware splitting based on subjects
- avoiding data leakage.

The notebook separates epochs based on subjects into training, validation, and testing sets, rather than choosing randomly:

- training
- validation
- testing

```
Split → Train: 8 | Val: 1 | Test: 1 subjects
Scaler fitted on 10,240 train epochs

Train: 10,160 | Val: 1,309 | Test: 1,189 sequences
```

The reason why it is so vital

A purely random epoch-based split leads to data leakage because:

some of the epochs from the same patients could be on test and train; model can simply memorize patient-specific features

This results in:

- unreasonably accurate results
- ineffective real-world model

The subject-based split prevents this from occurring.

Further Enhancements

Experiments With Spectrograms

This notebook uses:

- spectrogram processing

which means that:

- time-domain signals for EOG are converted to a time-frequency image.

The goal here was to better capture the relationship between frequency and sleep stage and thus enhance the extraction features from the EOG data.

Training Robustness

In addition, the notebook continues the efforts to stabilize training by using:

- ReduceLROnPlateau
- dropout
- CNN+LSTM structure

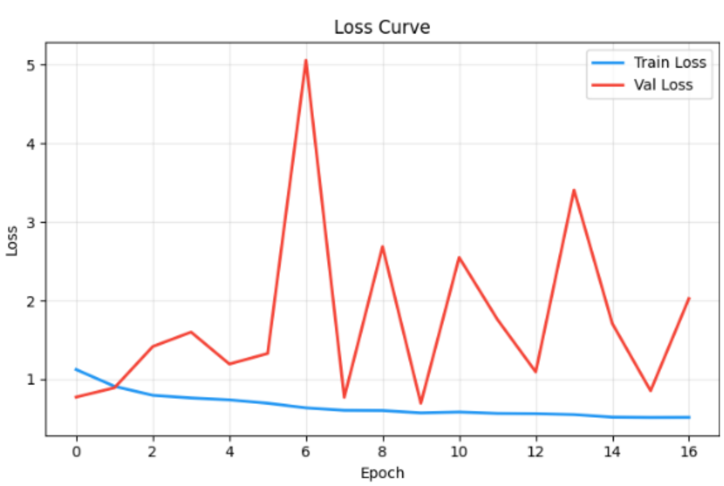
The contribution this notebook made is in a superior:

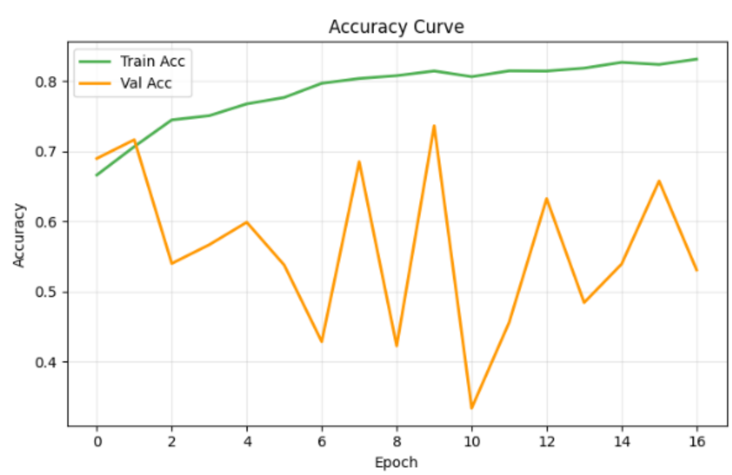
- experimental fairness
- generalizability of the model
- accuracy of the evaluation

CLASSIFICATION REPORT:

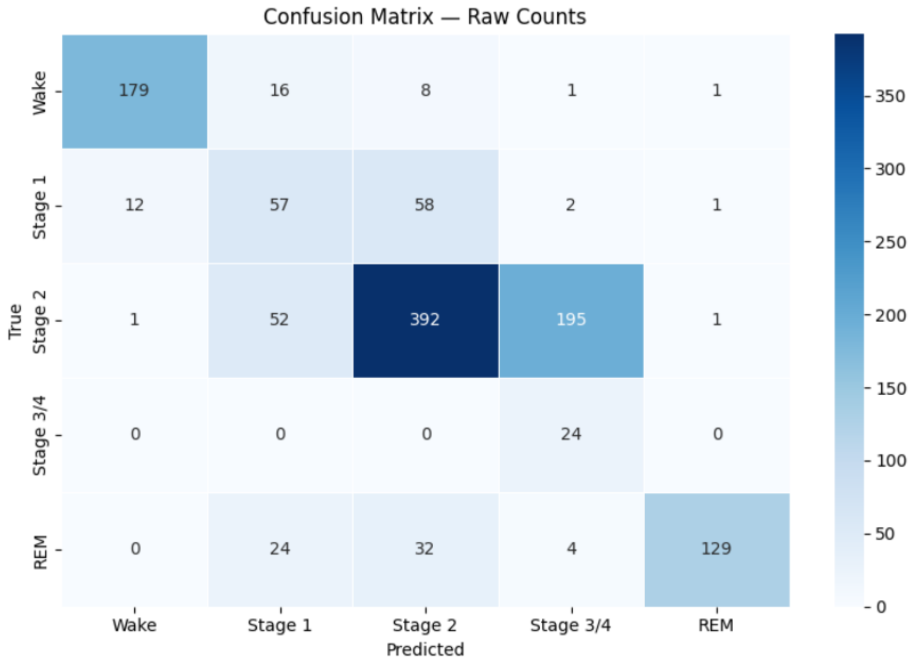
```
              precision    recall  f1-score   support

        Wake       0.93      0.87      0.90       205
     Stage 1       0.38      0.44      0.41       130
     Stage 2       0.80      0.61      0.69       641
   Stage 3/4       0.11      1.00      0.19        24
         REM       0.98      0.68      0.80       189

    accuracy                           0.66      1189
   macro avg       0.64      0.72      0.60      1189
weighted avg       0.79      0.66      0.71      1189

Macro F1    : 0.5999
Cohen Kappa : 0.5199
```

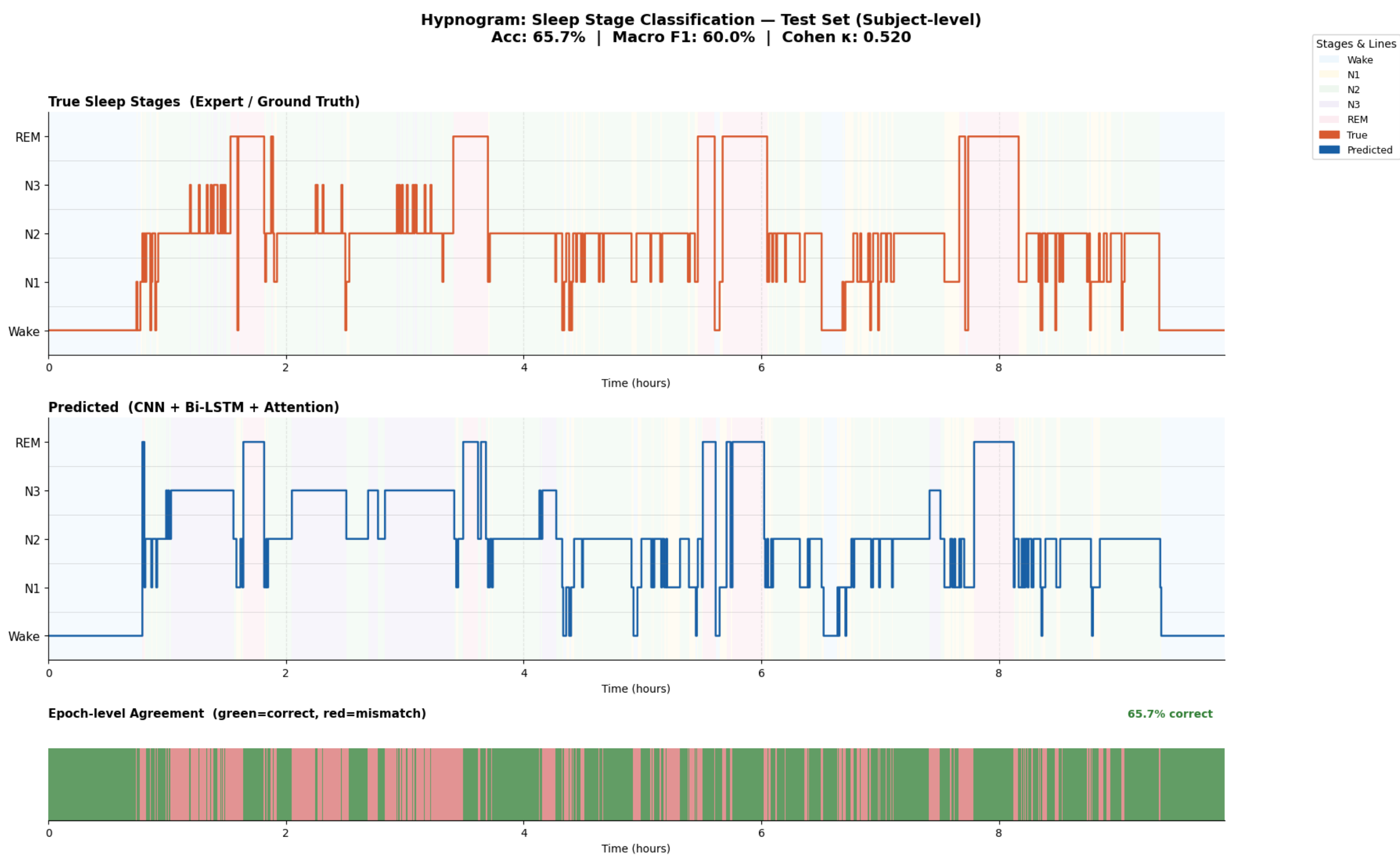
Curves:





Confusion Matrix & Hypnogram





2. P_EOG-R-ResNet1D

Key differences to earlier versions:

This notebook included more sophisticated techniques of imbalance handling and more improved training schemes.

New entries:

1. WeightedRandomSampler

One key inclusion was:

- WeightedRandomSampler

Its purpose:

- Deal with class imbalance
- Sample minority stages more frequently
- Less emphasize on majority stages

Reason behind it's importance:

Sleep staging data is inherently imbalanced.

Example of imbalance:

- Dominating class is N2
- Underrepresented class is N1

When no sample weightage is implemented, the model:

- Predominantly predicts the majority stages
- Yields poor recall of minority stages

2. Focal Loss

Inclusion of:

- FocalLoss

Its purpose:

- Prioritize training on hard examples
- Downplay the impact of easy majority samples

Its significance:

Focal Loss is known to enhance

- Minority class identification
- Robustness to class imbalance
- Training stability

Especially useful when it comes to:

- Classification of N1
- Dealing with awkward transition points

3. Stronger Regularization

The model had:

- Dropout layers

Its purpose:

- Prevent overfitting
- Enhance generalization

4. Pretrained ResNet1D-LSTM Model

A critical addition in this stage:

- pretrained ResNet1D-LSTM

Its significance:

Combined deep feature learning for EOG signal using ResNet1D with sequence learning through an LSTM model

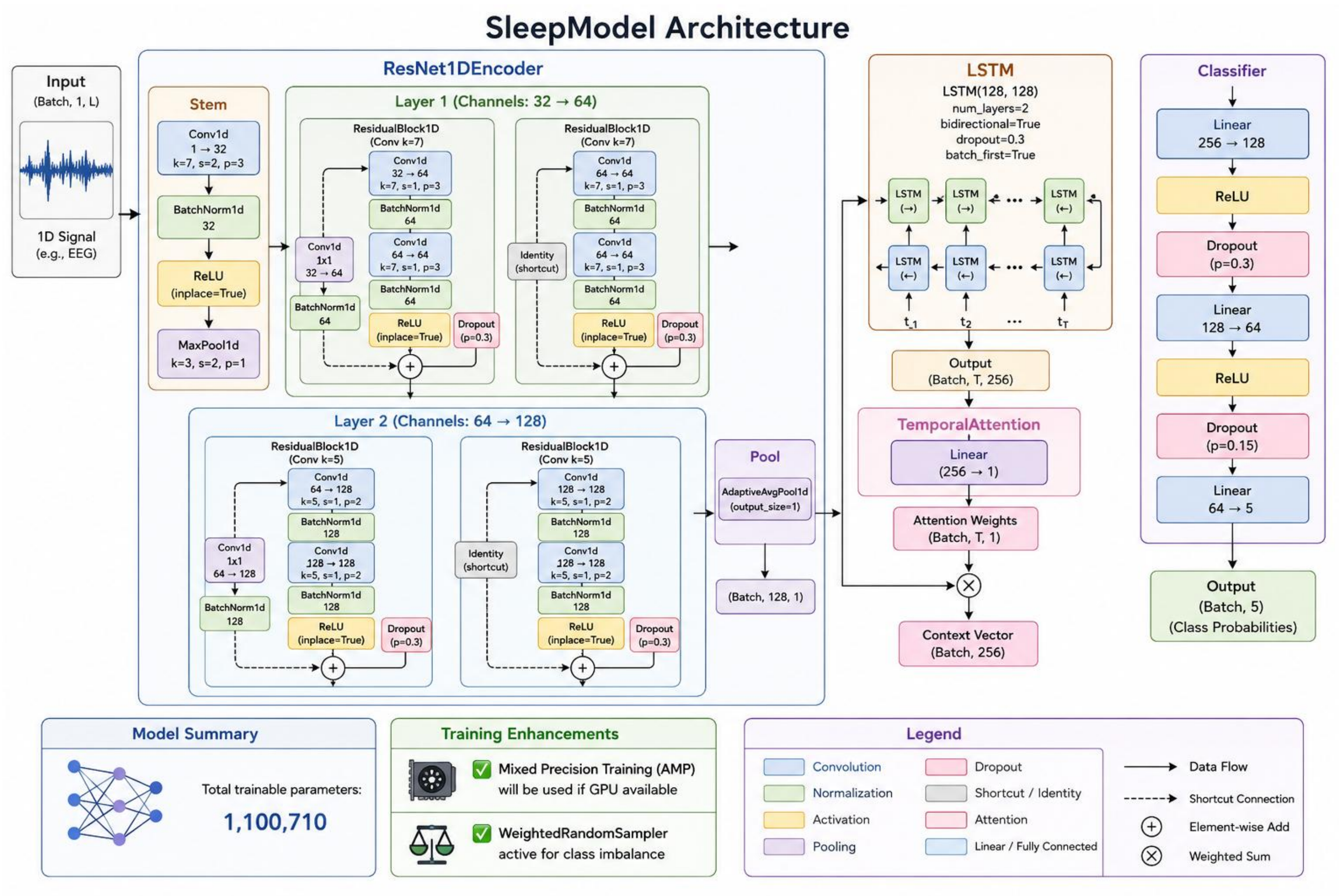
Reason behind its importance

Pretrained ResNet1D backbone leads to:

- Quality feature learning
- Faster convergence
- Better representation learning
- Reduced manual feature design needed

LSTM component to:

- Effectively model sequences of sleep epochs
- Predict transitions between sleep stages



Why ResNet1D-LSTM is a good choice:

As compared to simple CNN models, ResNet1D-LSTM provides:

- More sophisticated hierarchical feature learning
- More temporal analysis capacity
- Enhanced generalization capability
- Greater stability with challenging sleep patterns

Summary of this stage:

In this notebook, the primary goals were:

- Class imbalance handling

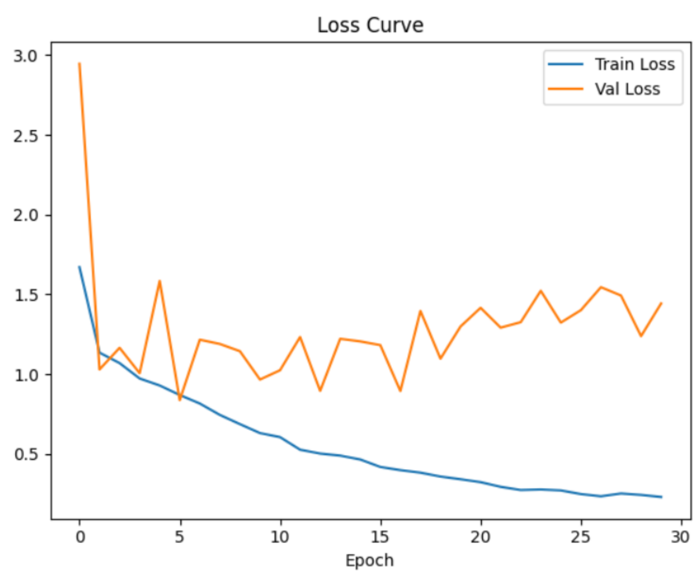
- Improved prediction for minority stages
- Application of pre-trained deeper models
- Modeling sequences of epochs
- Robustness in training

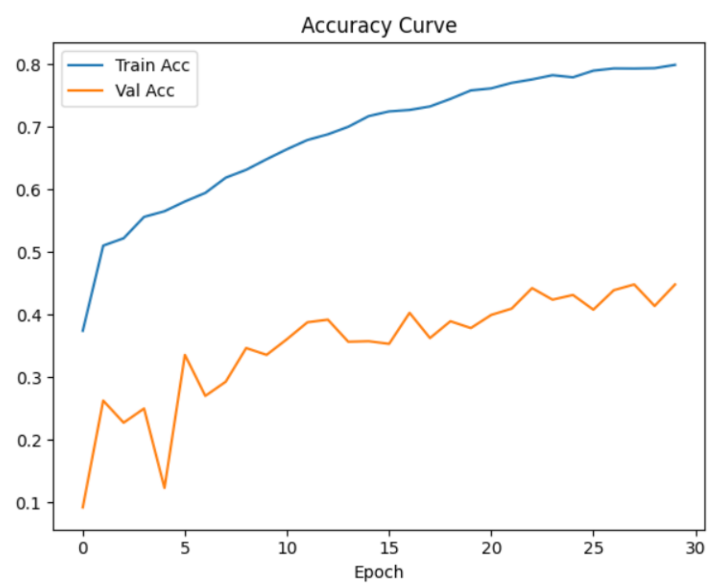
Classification Report

	precision	recall	f1-score	support
Wake	1.00	0.68	0.81	475
Stage 1	0.15	0.49	0.23	70
Stage 2	0.89	0.61	0.72	373
Stage 3/4	0.47	1.00	0.64	62
REM	0.54	0.70	0.61	149
accuracy			0.67	1129
macro avg	0.61	0.70	0.60	1129
weighted avg	0.82	0.67	0.71	1129

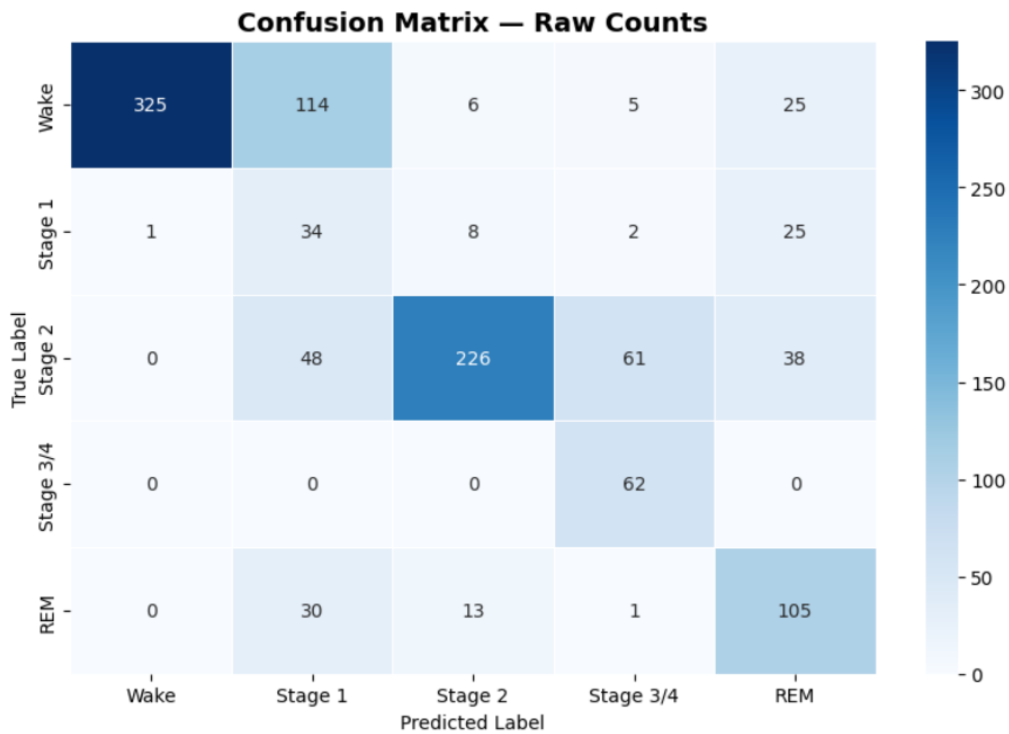
```
Macro F1  : 0.604
Cohen Kappa: 0.5624
```

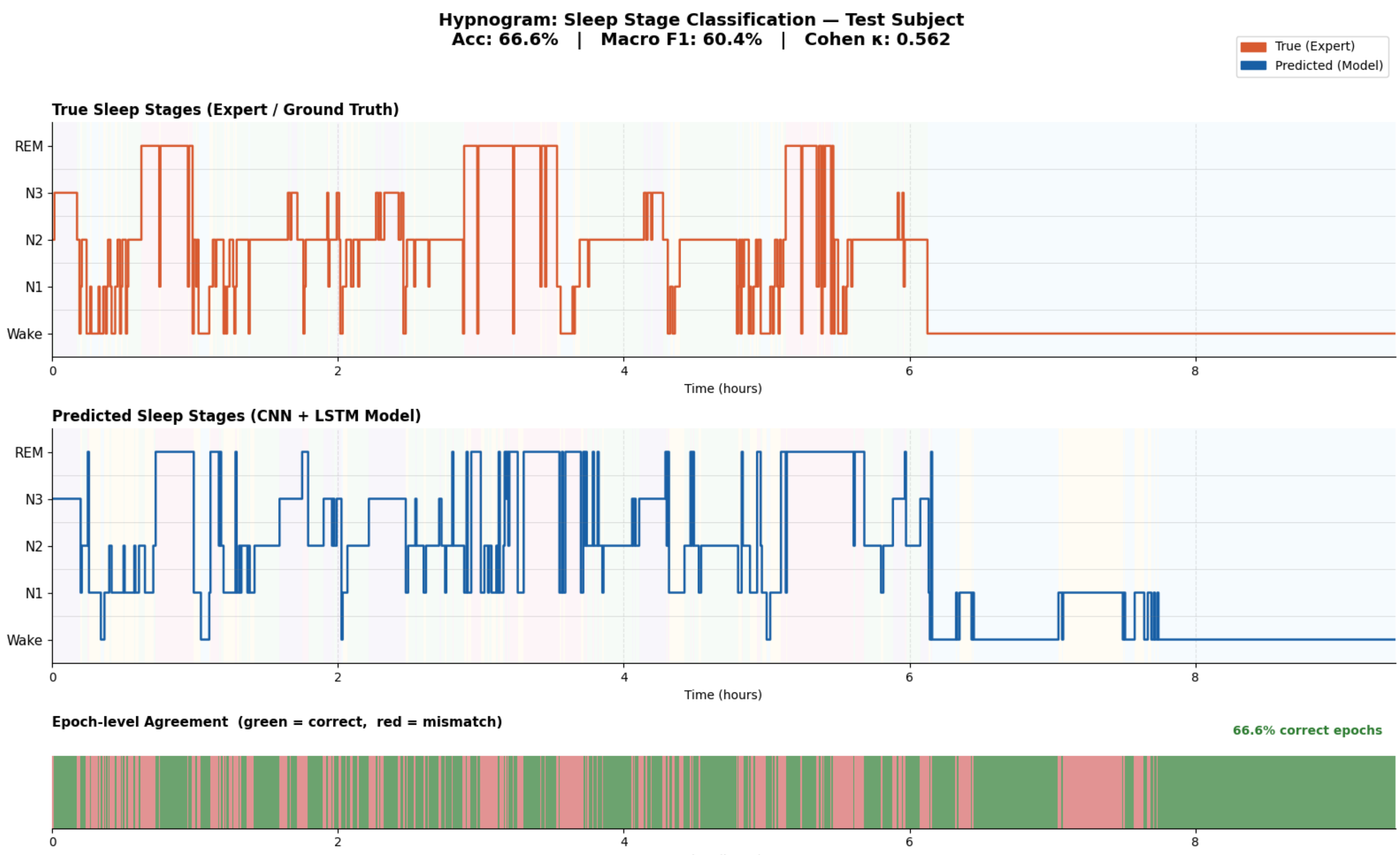
Curves





Confusion Matrix & Hypnogram





3. P_EOG-R-Final

Final Optimized Version

The final clean and optimized pipeline is the most likely.

Main features

- CNN Encoder + LSTM architecture
- Subject-split data splitting
- StandardScaler preprocessing
- Scheduler improvement
- Better organized training
- Better evaluation routine

Improvements in the Final Version

1. Cleaner pipeline

The data flow is more structured as:

- Preprocessing
- Data split
- Train
- Validate
- Test
- Metric
- Visualizations

2. Consistent training settings

The notebook maintains:

- ReduceLROnPlateau
- Dropout
- CrossEntropyLoss

which implies stability is preferred over exploring performance for the sake of getting better values.

3. More realistic experiments

At this stage, the project most likely has obtained:

- A more representative assessment
- Reduced data leakage
- Higher degree of generalization
- More robust system

Main objectives of the Final Version

The purpose of the last notebook is to get:

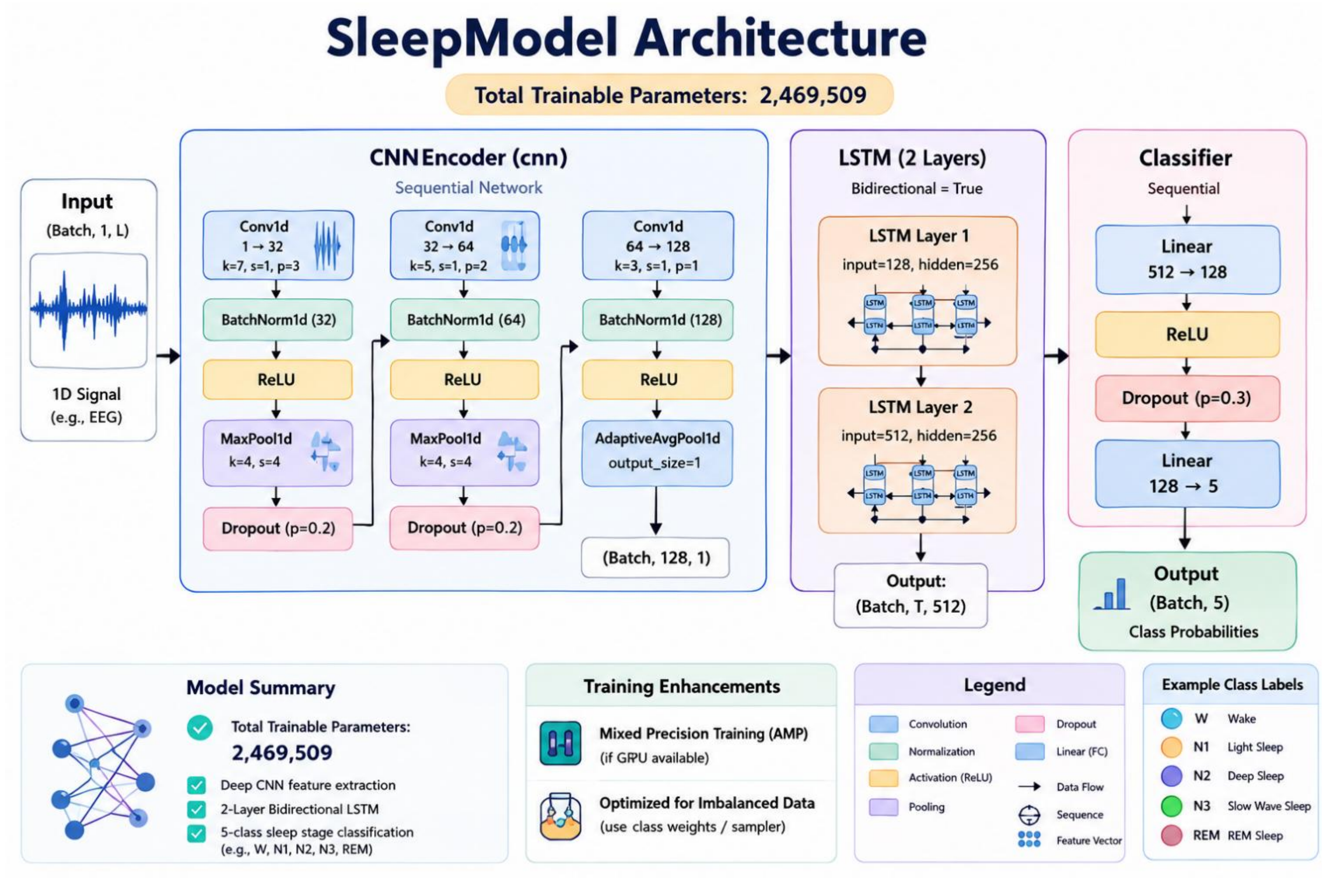
- A robust, reproducible output
- A well-organized report and presentation preparation stage
- A more robust output

Split

```
Train: 7 subjects → 9271 sequences
Val  : 2   subjects → 2258 sequences
Test : 1  subject  → 1129 sequences
```

Train: 70% - Val: 20% - Test: 10%

Final Model

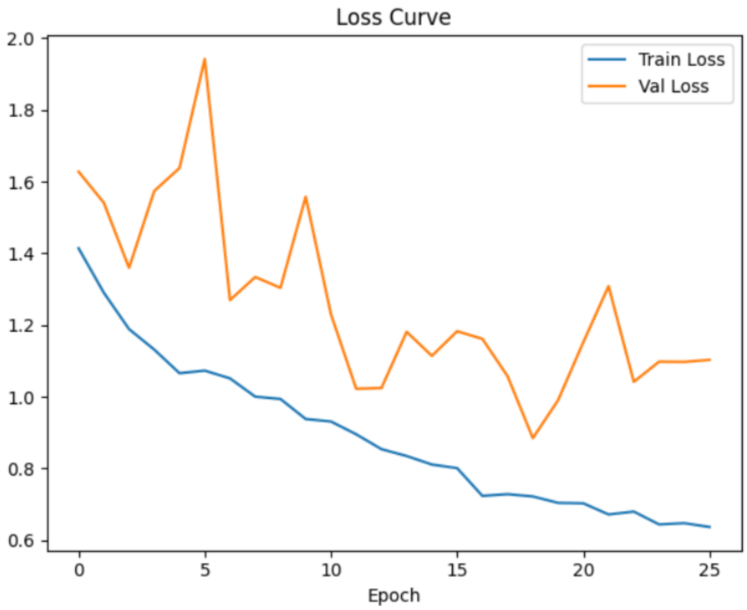


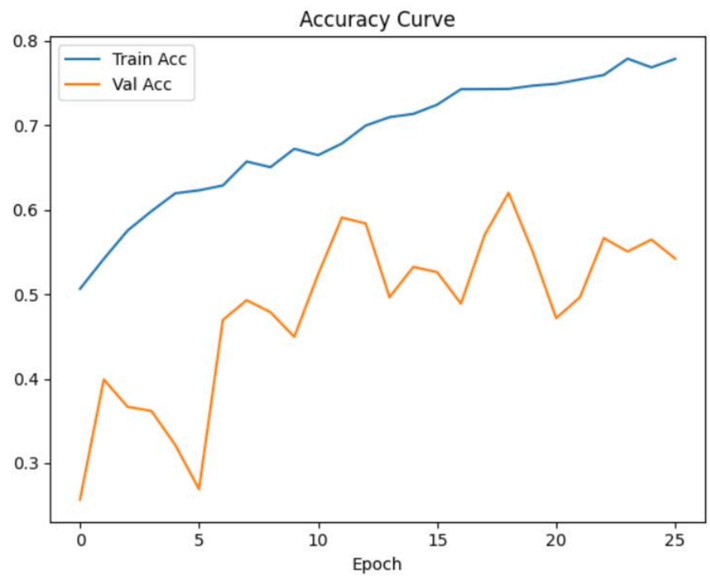
Classification Report

	precision	recall	f1-score	support
Wake	1.00	0.84	0.91	475
Stage 1	0.36	0.44	0.39	70
Stage 2	0.85	0.79	0.82	373
Stage 3/4	0.49	1.00	0.66	62
REM	0.66	0.74	0.70	149
accuracy			0.80	1129
macro avg	0.67	0.76	0.70	1129
weighted avg	0.84	0.80	0.81	1129

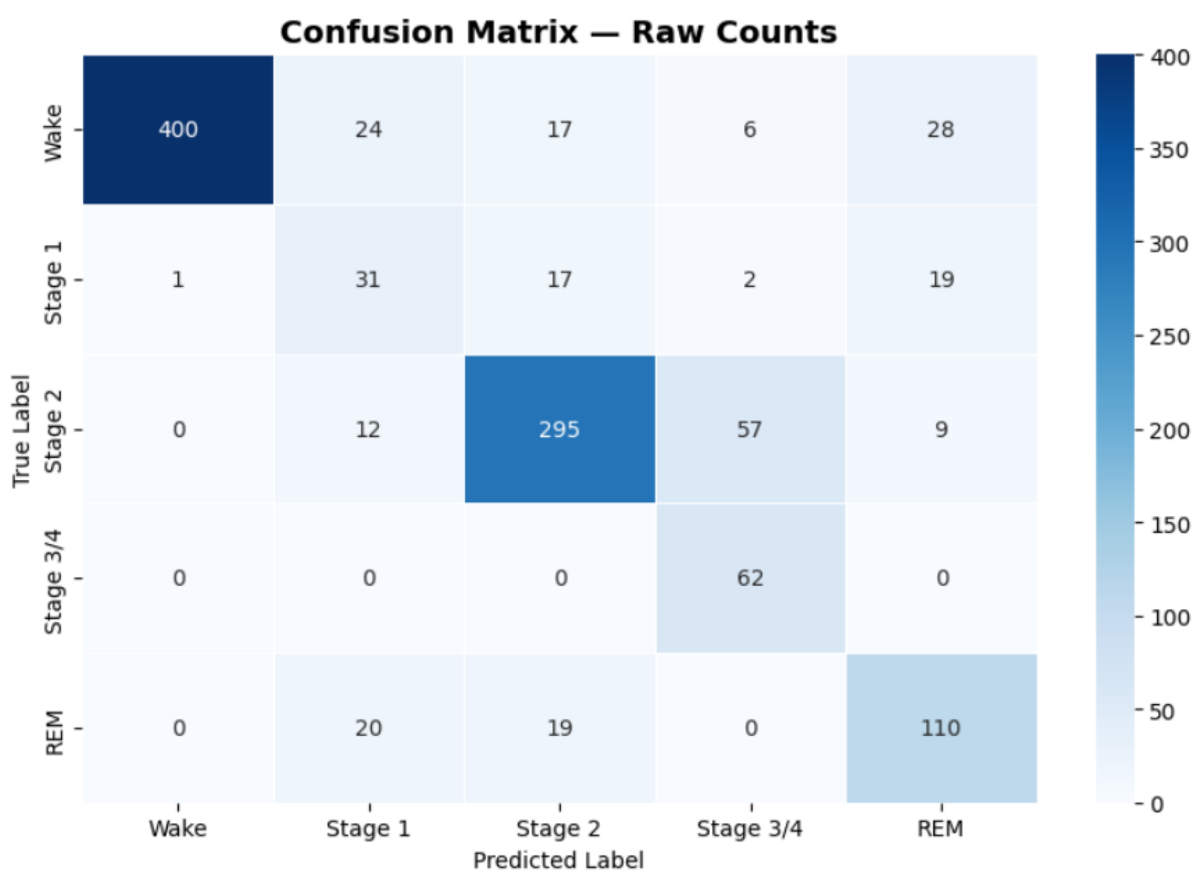
```
Macro F1  : 0.6962
Cohen Kappa: 0.7152
```

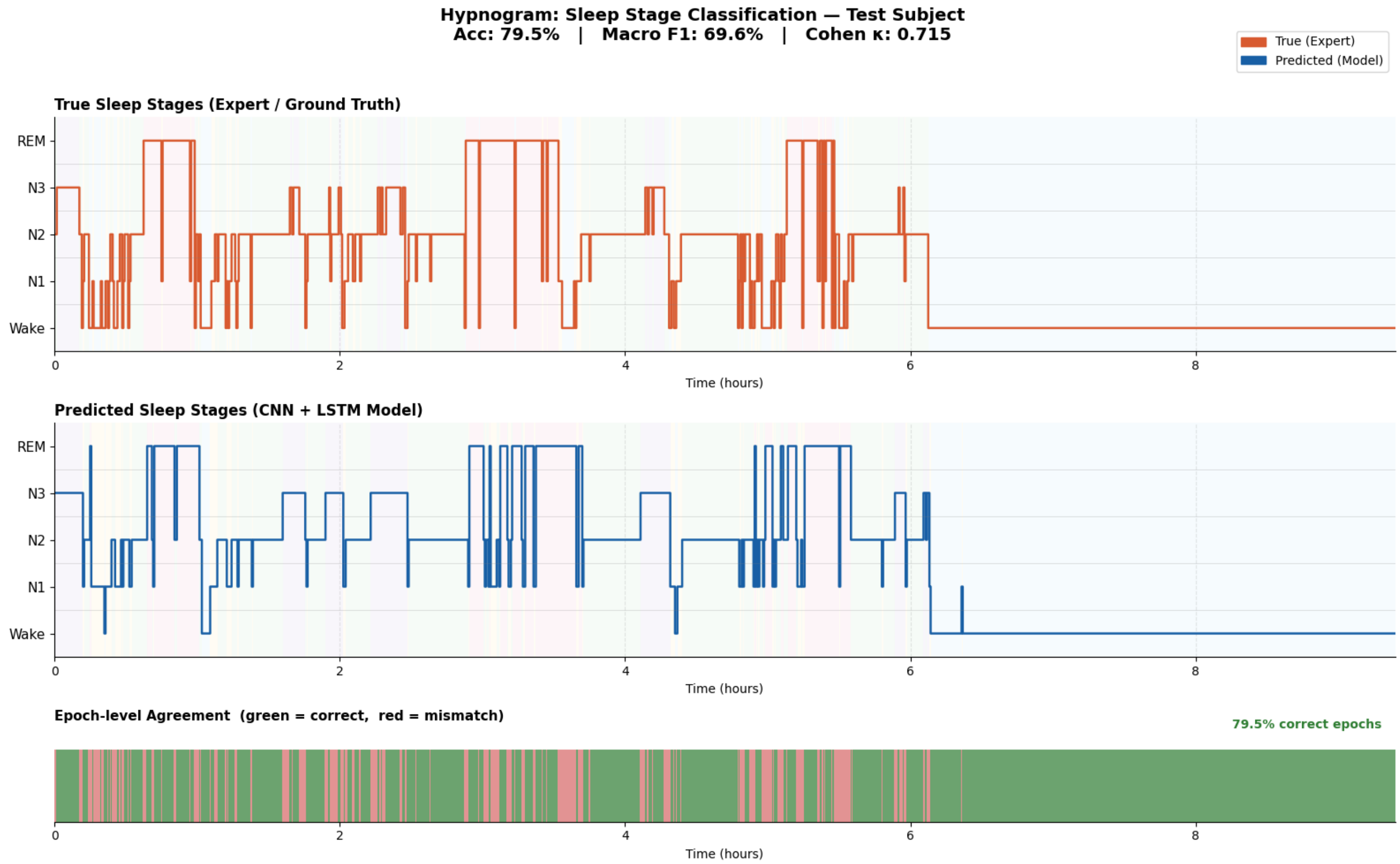
Curves





Confusion Matrix & Hypnogram





Technical advances in major revisions

Enhancement	Use Case
CNN + LSTM	Extract spatial+temporal features
Subject split	Avoid data leakage
WeightedRandomSampler	To deal with imbalance
Focal Loss	Enhance minority-class learning
Dropout	Preventing Overfitting.
ReduceLROnPlateau	Ensure training remains stable
Spectrograms	Time-frequency feature extraction
StandardScaler	Normalization of signal
F1-score & Kappa	Accuracy alone is a better measure

6th Conclusion:

To conclude, this project had proved that sleep stages could be accurately classified using solely EOG signals, instead of utilizing EEG channels. Throughout the project, several deep learning architectures and preprocessing methods were tested in order to better the performance of the classification models, as well as their robustness. The usage of the MESA Sleep Dataset allowed for analysis of real physiological sleep signal, as well as providing a base for analyzing EOG-only sleep classification models.

The project went through a number of stages, ranging from a simple CNN+LSTM network, through to highly tuned deep learning models. Several of the innovations developed within the project, such as subject-wise splitting, spectrogram images and WeightedRandomSampler, Focal loss, Dropout, and the ResNet1D-LSTM model, improved the generalization capabilities of the models and addressed issues with data leakage and class imbalance. The aforementioned techniques had successfully improved classification performance in minority classes such as N1 and REM.

The final and tuned model reached near 80% classification accuracy, with high Cohen's Kappa and F1-score values, illustrating reliable agreement between the classified and real sleep stages. The results successfully demonstrated the effectiveness of the physiological signals found within EOG data for classifying sleep stages.

Overall, this project has highlighted the potential use of deep learning and signal processing for cheaper, more accessible and less intrusive sleep monitoring systems. Potential areas for further development of this project include the use of a wider range of subjects, inclusion of other physiological signals, exploring transformer-based network structures and analysis with large, heterogeneous datasets to evaluate clinical applicability.

7th GitHub:

https://github.com/Usama-Mohammed/Sleep-Stage-Classification.git

8th Reference:

https://www.medrxiv.org/content/10.1101/2024.08.02.24311417v1

https://sleepdata.org/datasets/mesa?utm_m